Chest X-Ray Generation with a Spatial VAE

ECE 285 Homework 1

Abstract

A Gaussian VAE generates 128×128 chest X-rays using L1/SSIM reconstruction and KL regularization. Evaluation gives FID 362.72 and IS 1.8698; fidelity remains limited.

1 Data and model

Kaggle v2 [3]: 5,216 train/624 test images; labels and 16 validation images unused. Grayscale inputs: aspect-preserving resize, 128^2 center crop, range $[0, 1]$. Residual Gaussian heads [1]: 128 global and $4 \times 16 \times 16$ local coordinates. Decoder context learns $p_\theta(a \mid g)$, injected through a 1×1 projection. Generation: 1,152 independent normal-noise inputs; 5,950,481 parameters.

2 Pure-VAE objective and optimization

Warm start: 20-epoch VAE plus 5,000 pure-VAE updates; new spatial heads and fresh AdamW. Whitening q_a against $p_\theta(a \mid g)$ gives $q(g, \epsilon \mid x) = q_g(g \mid x) q_\epsilon(\epsilon \mid g, x)$, conditional rather than jointly diagonal. Reconstruction and KL share sampled g. The batch mean of this per-image loss is optimized:

$$R(x, \hat{x}) = \|x - \hat{x}\|_1 + \lambda_t P[1 - \mathrm{SSIM}(x, \hat{x})], \tag{1}$$

$$\mathcal{L}(x) = R(x, \hat{x}) + \beta_t K(x). \tag{2}$$

$$K(x) = D_{\mathrm{KL}}(q_g \| \mathcal{N}(0, I)) + E_{q(g|x)} D_{\mathrm{KL}}(q_a \| p_\theta(a \mid g)).$$

Expectation uses one sampled global code. SSIM [2]: 11×11 Gaussian window ($\sigma = 1.5$), valid convolution, $P = 128^2$. This heuristic reconstruction is not a likelihood ELBO. Generation draws $z \sim \mathcal{N}(0, I)$.

Batch 16, seed 285, 3,000 updates; all training images, no augmentation. $\beta = 0.1$. λ_t: 0 to 0.2 over 200 updates. AdamW $(0.9, 0.999)$, decay 10^{-4}; cosine base 10^{-4}, floor 10^{-5}, global horizon 3,000; clip 10.

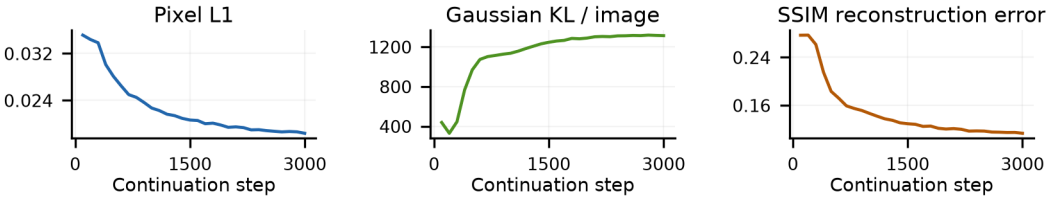


Figure 1: Recorded pixel L1, raw KL, and SSIM reconstruction error interval means. The spatial counter restarts at 0; its conditional 1,152-D KL differs from the earlier 128-D KL.

3 Generation evaluation

Ignite [4] uses Torchvision Inception-v3 `IMAGENET1K_V1`, input transform enabled: grayscale to RGB, antialiased 299^2 resize, ImageNet normalization. FID uses 2,048-D pool features; IS uses 1,000-class probabilities. Both use 512 identical test images and 512 prior samples (seed 285); prior dimensions 128/1,152 and latent vectors differ.

Pure VAE	FID ↓	IS ↑
Original 20-epoch checkpoint	376.38	1.6890
Conditional spatial VAE	362.72	1.8698

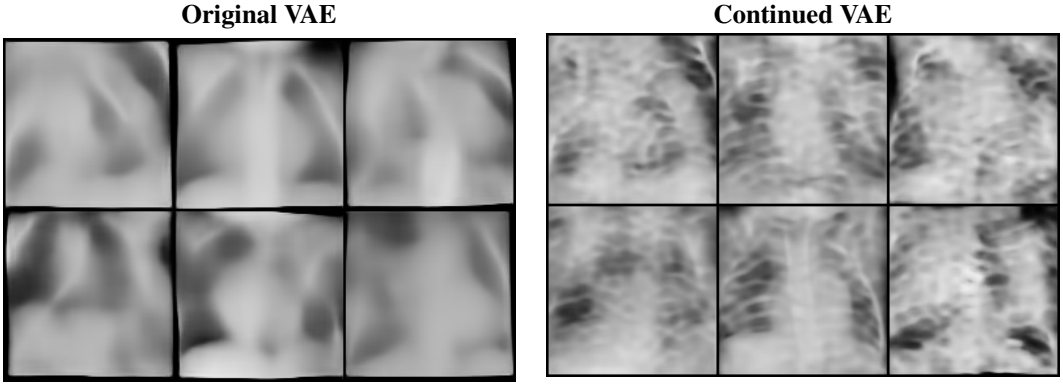


Figure 2: First six prior samples in fixed row-major order, seed 285, temperature 1. The fixed seed is shared; model-specific prior dimensions differ. No individual image selection or sharpening. Three complete 25-sample grids accompany the code.

4 Interpretation and limitations

Generated samples have clearer thoracic outlines and visible rib bands than the original VAE. Soft regions, irregular rib continuity and mottled texture remain. The modest FID decrease alone does not establish realistic anatomy. Losses, augmentation, and budget change together, confounding causal attribution. Spatial capacity, KL dimensionality, and posterior factorization also change. Reconstruction monitors are train-seen, including through the initializer; test images were observed during development.

One seed and 512 samples limit statistical conclusions. This Torchvision/Ignite protocol is not numerically interchangeable with TensorFlow-FID or Clean-FID. ImageNet features and visual plausibility do not establish anatomical correctness. Verified patient-separated validation, repeated runs, and matched-budget loss comparisons remain future work.

References

[1] D. P. Kingma and M. Welling. Auto-Encoding Variational Bayes. ICLR, 2014.

[2] Z. Wang et al. Image quality assessment: From error visibility to structural similarity. IEEE Trans. Image Process., 2004.

[3] P. Mooney. Chest X-Ray Images (Pneumonia). Kaggle, version 2. https://www.kaggle.com/datasets/paultimothymooney/chest-xray-pneumonia.

[4] PyTorch-Ignite. FID and Inception Score evaluation. https://pytorch-ignite.ai/blog/gan-evaluation-with-fid-and-is/.